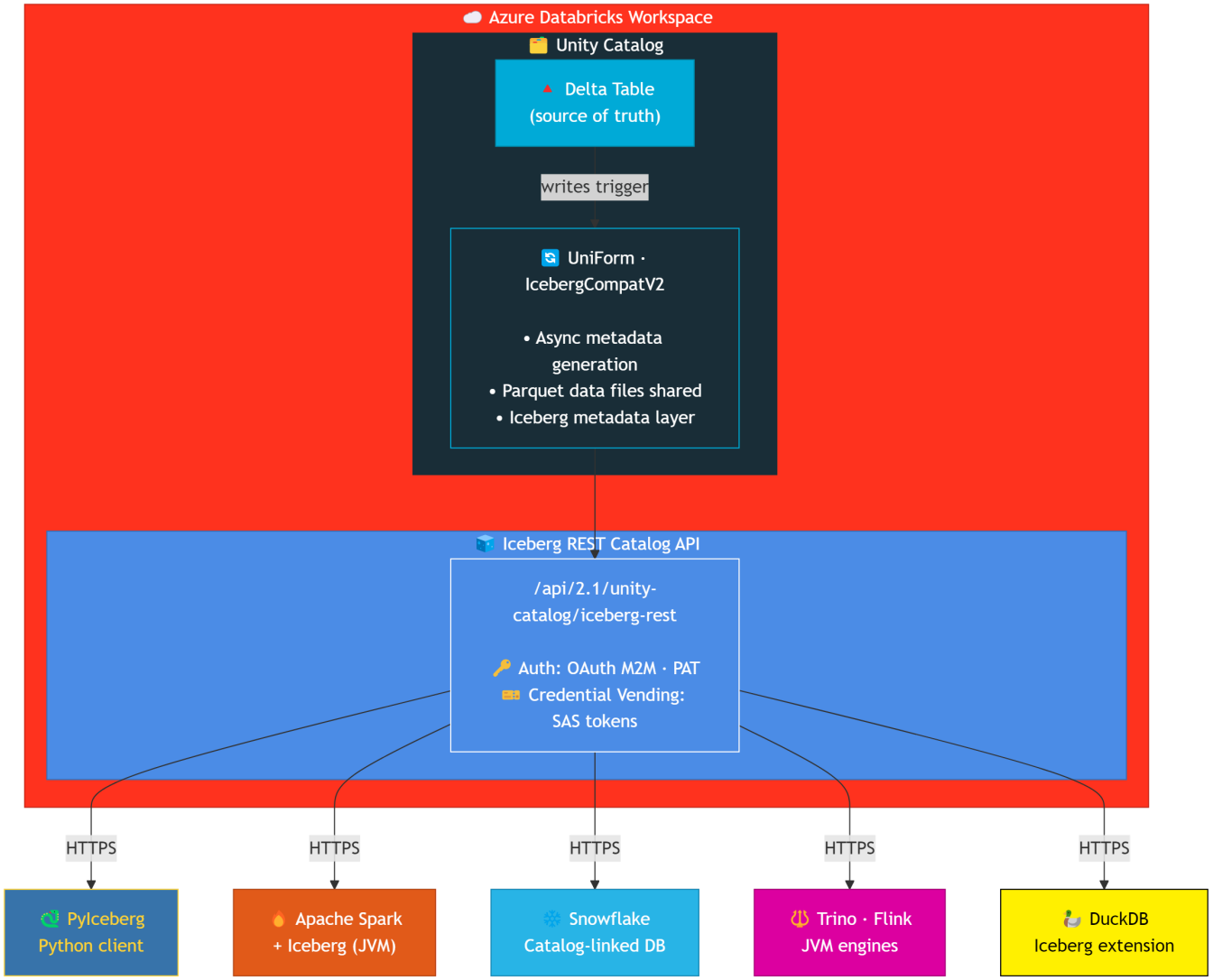
Closure Note — Investigate Iceberg REST Catalog API Feasibility

ADO Ticket: #186438 **Area**: newUM\Data Team **Sprint**: TBD **Priority**: P2 **Sources**: All claims verified against official Microsoft Learn docs (fetched 2026-03-23). See References for full list. **Project context**: `clients/oncohealth/knowledge.json` v1.7.0 — operational facts cited as [K].

Architecture Overview

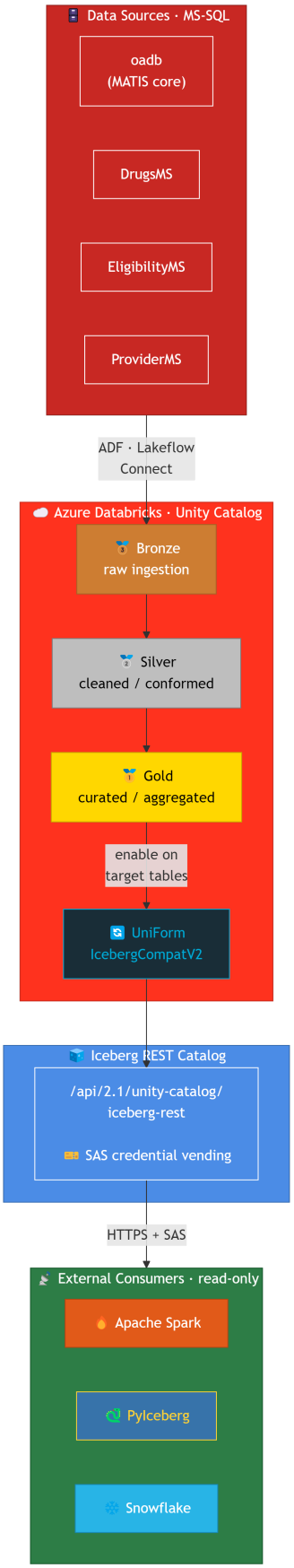


▶ Mermaid source (click to expand)

Data flow: Writes go through Databricks (Delta) → UniForm generates Iceberg metadata async → External clients read via REST Catalog API → Credential vending provides temporary ADLS SAS tokens.

> *Sources: [S1] endpoint + credential vending; [S2] UniForm async metadata generation.*

Applied to newUM



▶ Mermaid source (click to expand)

The above architecture maps to our confirmed environment [K: `tech_stack.data`, `environments.databricks`]:

Layer	newUM Reality	Source
Delta tables	Unity Catalog with Bronze/Silver/Gold medallion	[K: `tech_stack.data`]
Workspace (DEV)	`https://adb-2393860672770324.4.azuredatabricks.net/` — network-blocked for our account	[K: `databricks.workspaces[0]`]

Workspace (TEST)	URL unknown — access GRANTED via ticket #0035611 (2026-03-23)	[K: `access_inventory.results["Databricks test"]`]
Workspace (UAT/PROD)	URLs unknown	[K: `databricks.workspaces[2-3]`]
PAT token	`visualstudio-carlos` created, expires 2027-03-22	[K: `access_inventory.results["Databricks test"]`]
Auth method	Entra ID button click required (Okta SSO does NOT auto-login to Databricks)	[K: `access_status.access_inventory.note`]
Data Team Lead	Michal Mucha — runs daily standups, Databricks Lakeflow Connect chat	[K: `key_contacts["Michal Mucha"]`]
DevOps contact	`devopsrequest@oncologyanalytics.com`	[K: `onboarding_documents.devops_email`]
Target databases	oadb (main), DrugsMS, EligibilityMS, ProviderMS — all MS-SQL, candidates for lakehouse migration	[K: `environments.databases`]

Cost Estimation

Component	Cost Impact
Iceberg metadata generation	Runs on same compute as Delta writes — marginal increase in driver resource usage
Storage	Iceberg metadata files stored alongside Delta metadata — negligible (~KB per version)
API calls	REST Catalog API calls — included in Databricks pricing, no extra charge
Service principal	No additional licensing cost
Network	If Private Link required — Azure Private Link charges apply
Total incremental cost	**Near-zero** — no new compute or storage required

> ***Note***: *Cost estimation is inferred from architecture described in [S2]. Databricks docs state metadata generation "might increase the driver resource usage" [S2] but provide no cost figures. For high-throughput write workloads, driver memory impact should be validated in TEST.*

Investigation Files

- Full report: `clients/oncohealth/tickets/186438-iceberg-rest-catalog/output.md`

Pros

- **Zero data duplication** — Iceberg reads use same Parquet files as Delta; only metadata is generated ([S2]: "*A single copy of the data files serves multiple formats*")
- **Native Azure Databricks support** — endpoint is built-in, no external infrastructure needed ([S1]: "*Unity Catalog provides an implementation of the Iceberg REST catalog API*")
- **Broad client compatibility** — Pyiceberg, Spark, Snowflake, Trino, Flink supported per [S1]; DuckDB via its own Iceberg extension (not in Databricks docs)
- **Credential vending** — temporary SAS tokens issued automatically, default 1h expiry ([S1]: "*The default expiration time is one hour*")
- **Low operational overhead** — metadata generation is automatic and async ([S2]: "*asynchronously after a Delta Lake write transaction completes*")
- **Standards-based** — uses official Apache Iceberg REST Catalog spec ([S1])
- **Public Preview (DBR 16.4+)** — on GA track, production-viable with preview caveats ([S1]: "*Public Preview in Databricks Runtime 16.4 LTS and above*")

Cons

- **Read-only for Delta+UniForm tables** — external clients cannot write; writes must go through Databricks ([S2]: "*Iceberg client support is read-only. Writes are not supported.*")
- **Metadata lag** — Iceberg metadata generated asynchronously; may lag behind latest Delta version ([S2]: "*Delta table versions do not align with Iceberg versions*")
- **Protocol upgrade partially irreversible** — Iceberg reads can be disabled by unsetting `delta.universalFormat.enabledFormats`, but Delta protocol version upgrades and column mapping **cannot** be undone ([S2]:

"*You can turn off Iceberg reads by unsetting the delta.universalFormat.enabledFormats table property. Upgrades to Delta Lake reader and writer protocol versions cannot be undone.*")

- **Deletion vectors incompatible with Iceberg v2** — tables need `REORG` before enabling; however, **Iceberg v3 supports deletion vectors** ([S2]: "*Apache Iceberg v3 supports deletion vectors*")
- **Public Preview status** — not yet GA; breaking changes possible (low risk given timeline) ([S1])
- **Snowflake+Entra requires public networking** — cannot use Private Link for Entra OAuth ([S1]: "*must use public networking when authenticating with an Entra service principal*")

Risks & Open Questions

#	Risk/Question	Severity	Mitigation
1	**No workspace access validated** — DEV is `adb-2393860672770324` but network-blocked; TEST URL unknown despite access granted [K: `databricks.workspaces`, `access_inventory`]	HIGH	Reply to DevOps ticket #0035611 requesting TEST workspace URL
2	**Metadata staleness** — Iceberg metadata may lag Delta writes	MEDIUM	Monitor `converted_delta_version`; use `MSCK REPAIR TABLE` if needed
3	**Target tables unknown** — which Unity Catalog tables need UniForm? Bronze/Silver/Gold layers exist [K: `tech_stack.data`] but specific tables not yet identified	MEDIUM	Coordinate with Michal Mucha [K: `key_contacts`] for table selection
4	**Network restrictions** — DEV workspace already network-blocked for us [K: `access_inventory`]; TEST/UAT/PROD may have similar restrictions	MEDIUM	Validate firewall/VNet/Private Link config; contact DevOps (`devopsrequest@oncologyanalytics.com` [K])
5	**External data access not enabled** — metastore may need admin config	MEDIUM	Requires UC Admin to enable; escalate via Erik Hjortshoj (SVP Engineering) [K: `key_contacts`]
6	**Protocol upgrade partially irreversible** — Iceberg reads can be toggled off, but Delta protocol versions and column mapping cannot be undone ([S2])	LOW	Test on non-prod table first; protocol is forward-compatible
7	**Deletion vectors on existing tables** — need REORG for Iceberg v2; Iceberg v3 supports them natively ([S2])	LOW	Schedule during maintenance window; REORG is idempotent
8	**Write via Iceberg not supported for Delta UniForm** — confirmed	INFO	Out of scope per ticket; writes stay in Databricks
9	**MFA blocks automation** — Databricks login requires Entra ID button click, not automated via Okta SSO like ADO/SharePoint [K: `access_inventory.note`]	LOW	Manual validation needed; user must be at keyboard
10	**Unresolved unknowns** — TEST/UAT/PROD URLs unknown (U2), preferred workspace unknown (U6) [K: `unknowns`]	MEDIUM	Resolve with Michal or DevOps before proceeding

Recommended Next Steps

1. **Obtain TEST workspace URL** — reply to DevOps ticket #0035611; contact `devopsrequest@oncologyanalytics.com` [K]
2. **Validate existing PAT** — token `visualstudio-carlos` (exp 2027-03-22) [K] should work for initial curl test against the Iceberg REST endpoint
3. **List UC catalogs/tables** — use PAT to query TEST workspace; identify Bronze/Silver/Gold tables [K: `tech_stack.data`] for UniForm candidates
4. **Coordinate with Michal Mucha** [K: `key_contacts`] — he leads Databricks Lakeflow Connect [K: `communication.channels[0].active_chats`]; align on which tables to expose
5. **POC on test table** — enable UniForm, validate PyIceberg read from external network
6. **Create service principal** — with `USE CATALOG`, `USE SCHEMA`, `SELECT`, `EXTERNAL USE SCHEMA`; request via DevOps [K]
7. **Enable external data access** on metastore — escalate to UC Admin via Erik Hjortshoj [K: `key_contacts`]
8. **Document network path** — confirm external service can reach workspace endpoint (DEV already blocked for us [K])
9. **Resolve unknowns U2 & U6** [K: `unknowns`] — TEST/UAT/PROD URLs and preferred workspace for initial validation

References

All claims in this document were verified against official Microsoft Learn documentation, fetched 2026-03-23:

ID	Title	URL	Last Updated
[S1]	Access Azure Databricks tables from Apache Iceberg clients	https://learn.microsoft.com/en-us/azure/databricks/external-access/iceberg	2026-03-19
[S2]	Read Delta tables with Iceberg clients (UniForm)	https://learn.microsoft.com/en-us/azure/databricks/delta/uniform	2026-03-06
[S3]	Enable external data access on the metastore	https://learn.microsoft.com/en-us/azure/databricks/external-access/admin#external-data-access	—
[S4]	Databricks service principals / Auth	https://learn.microsoft.com/en-us/azure/databricks/dev-tools/auth/	—
[S5]	Pyiceberg REST catalog configuration	https://py.iceberg.apache.org/configuration/#rest-catalog	—
[S6]	Apache Iceberg REST API spec	https://github.com/apache/iceberg/blob/master/open-api/rest-catalog-open-api.yaml	—
[K]	Project knowledge base	`clients/oncohealth/knowledge.json` v1.7.0	2026-03-23